NOTE

The contingency table consisting of m rows and n columns.
The observed data is presented in the form of a contingency table :

		Attribute B						Total
		B_1	B_2	…	B_j	…	B_n	
Attribute A	A_1	O_{11}	O_{12}	…	O_{1j}	…	O_{1n}	R_1
	A_2	O_{21}	O_{22}	…	O_{2j}	…	O_{2n}	R_2
	:	:	:	:	:	:	:	:
	.	.	.	.	.	.	.	.
	A_i	O_{i1}	O_{i2}	…	O_{ij}	…	O_{in}	R_i
	:	:	:	:	:	:	:	:
	.	.	.	.	.	.	.	.
	A_m	O_{m1}	O_{m2}	…	O_{mj}	…	O_{mn}	R_m
Total	C_1	C_2	…	C_j	…	C_n	$N = m \times n$	

Step 3 : Level of significance

Fix the desired level of significance α

Step 4 : Calculation

Calculate the value of the test statistic as

$$\chi_0^2 = \sum_{i=1}^{m}\sum_{j=1}^{n}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$$

Step 5 : Critical value

The critical value is obtained from the table of χ^2 with $(m-1)(n-1)$ degrees of freedom at given level of significance, α as $\chi^2_{(m-1)(n-1),\alpha}$.

Step 6 : Decision

Decide on rejecting or not rejecting the null hypothesis by comparing the calculated value of the test statistic with the table value. If $\chi_0^2 \geq \chi^2_{(m-1)(n-1),\alpha}$ reject H_0.

Note:

- N, the total frequency should be reasonably large, say greater than 50.
- No theoretical cell-frequency should be less than 5. If cell frequencies are less than 5, then it should be grouped such that the total frequency is made greater than 5 with the preceding or succeeding cell.

Example 2.9

The following table gives the performance of 500 students classified according to age in a computer test. Test whether the attributes age and performance are independent at 5% of significance.

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